Toolkit from the
US-ASEAN Smart Cities Mobility Program

TRANSIT SERVICE PLANNING FOR SUSTAINABLE TOURISM TRAVEL

INSIGHTS FROM PHUKET AND LAS VEGAS

INTRODUCTION

Tourism places unique and sometimes competing demands on urban transportation systems. Visitors, residents, and workers travel at different times, use different modes, and rely on different payment tools; yet they share the same streets and transit networks. As tourism grows across ASEAN cities, these differences complicate the operation of transit services and efforts to deliver reliable, efficient, and sustainable mobility. This toolkit provides a structured approach for understanding and responding to those varied travel needs, helping cities assess tourism-related travel patterns and optimize transit services for both visitors and local users.

The toolkit documents best practices, research methods, and analytical approaches developed through the Smart Mobility Program, a U.S. Department of Transportation-led initiative under the U.S.-ASEAN Smart Cities Partnership. Drawing on peer-to-peer exchanges among eight cities in the United States and Southeast Asia, with research support from the METRANS Transportation Consortium at the University of Southern California, it distills lessons from the Las Vegas–Phuket pairing. Together, these partners examined how transit systems can better serve tourist-heavy corridors while maintaining accessibility, efficiency, and quality of service for residents and workers. The partnership included participation from the following agencies:

Digital Economy and Promotion Agency: DEPA supports Thailand's comprehensive digital transformation efforts. They currently support over 100 smart cities with programs in the areas of environment, mobility, energy, governance, economy, and more.

Box 1. What is the U.S. - ASEAN Smart Mobility Program?

This program addresses the transportation priorities of participating ASEAN smart cities by identifying innovative solutions to urban transportation challenges. The primary participants are local and regional transportation planner and policymakers from eight cities across the U.S. and Southeast Asia working on projects of mutual interest. The program focuses on how effective policy setting, good planning practices, the judicious use of technology, and targeted engagement with the private sector can increase mobility and improve transportation network operations. Participating cities include:

- Johor Bahru, Malaysia and Portland, Oregon
- Phuket, Thailand and Las Vegas, Nevada
- Phnom Penh, Cambodia and Boston, Massachusetts
- Jakarta, Indonesia and Los Angeles, California

The University Partnership Program (UPP) was formed to assist the city pairs with field research to support policy objective. The UPP is led by the METRANS Transportation Consortium at the University of Southern California and brings together four Southeast Asian Universities to lead research projects:

- Chulalongkorn University
- Institute of Technology of Cambodia
- Universitas Indonesia
- Univerisity Teknologi Malaysia

Regional Transportation Commission of Southern Nevada (RTC): RTC is a regional organization responsible for overseeing public transportation, traffic management, roadway design and construction, transportation planning, and regional planning across Southern Nevada.

Phuket and Las Vegas are both major tourism hubs facing the challenge of accommodating large, variable visitor flows within constrained urban transportation systems. Drawing on this shared context, the toolkit summarizes best practices and research aimed at advancing five core objectives common to tourism destinations: **reducing ground transportation congestion, improving safety, lowering emissions and environmental impacts, reducing travel costs for users, and controlling operating costs for providers**.

The toolkit focuses on increasing transit mode share, particularly shifting trips from private motorized modes to transit, as a primary strategy for achieving these objectives. While non-transit approaches can also support sustainable tourism mobility, the ASEAN context warrants a strong emphasis on transit-based solutions. Many ASEAN tourist cities are rapidly motorizing, with high shares of two- and three-wheel vehicles, while road networks are already operating at or near capacity. Opportunities for roadway expansion are limited, and where they exist, they often conflict with broader urban and equity goals.

Given these constraints, transit offers both a throughput advantage and a more equitable means of serving residents, workers, and visitors alike. In smaller and mid-sized ASEAN tourist destinations, this typically means bus transit, which remains the backbone of public transport systems. Accordingly, this toolkit is designed for practitioners from government, the private sector, transportation providers, and tourism officials from business and the public sector seeking practical methods to improve bus service delivery and increase transit use in tourism-intensive environments.

What you will learn from this toolkit

This toolkit documents how the Phuket–Las Vegas city pairing, in collaboration with research partners from Chulalongkorn University, approached the challenge of delivering reliable, safe, and clean transit service in Phuket. The focus is on strengthening and optimizing existing bus services rather than developing new lines. It, therefore, begins with building a clear understanding of the stakeholders involved in transit planning and operations. The toolkit provides a practical, step-by-step approach to assessing tourism-related travel demand and identifying opportunities to improve service. Through this process, you will learn the following:

1. The stakeholders and their roles in sustainable tourism travel in ASEAN nations. Government entities are typically responsible for planning and licensing routes, building infrastructure, and providing regulatory oversight, and, therefore, need reliable information on travel behavior, origins and destinations, and demand patterns. Transit service providers, usually private firms, must understand their customers to operate efficiently, innovate, and, in the absence of public subsidies, remain financially viable. At the same time, tourists and residents have their own profiles, needs, and desires. Effective planning, on both the public and private sector sides, requires a strong understanding of both groups of travelers.

2. Methods and tools for both transit planning. The toolkit includes a practical “how-to” primer on data-driven transportation planning, supported by templates and example materials. It presents a mixed-methods approach that combines travel surveys, used to inform route and infrastructure planning, with in-depth interviews and observational techniques designed to develop transit customer personas. Together, these methods generate actionable insights for both public agencies and transit providers.

3. Best practices and suggested roles for public and private actors. This toolkit will summarize approaches to coordinating the responsibilities of national ministries, local planning and transportation agencies, transit operators, industry and economic development groups, and multilateral development organizations. The focus is on clarifying roles and aligning incentives to support effective transit planning and service delivery in tourism-oriented ASEAN cities.

All of these lessons – covering stakeholder engagement, analytical methods, and institutional coordination – are grounded in an in-depth case study of sustainable tourism planning in Phuket. Insights from Las Vegas provide a comparative perspective that helps highlight transferable strategies and contextual differences.

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SUSTAINABLE TOURISM TRAVEL: DEFINING THE PROBLEM IN THE ASEAN CONTEXT

Tourism-oriented transit systems operate at the intersection of fragmented institutions, diverse user groups, and highly place-specific travel patterns. Understanding how these factors interact requires looking beyond abstract distinctions between tourists and residents or planners and operators, and examining how demand and supply are shaped by local context. The experiences of Phuket and Las Vegas illustrate how differences in arrival modes, information environments, regulatory structures, and urban form influence travel behavior and the attractiveness of transit. The following comparison highlights these contrasts and helps explain why tourism cities face distinct challenges in increasing transit use (Table 1 provides a summary overview of the two cities).

Las Vegas benefits from proximity to major population centers, most notably Southern California, which means a majority of visitors are domestic and arrive by private car. Once in the city, many tourists park their vehicles and travel on foot within the highly concentrated resort corridor, limiting demand for longer-distance transit trips. Las Vegas is a large, low-density metropolitan area of over two million residents, with a publicly operated transit system that provides extensive bus coverage and unified branding across the region. For both tourists and residents, transit must contend with high car ownership, abundant parking, and roadway designs that prioritize automobile access.

Phuket, by contrast, is a geographically isolated island destination where most tourists arrive by air, with a much larger share of international tourists, and immediately require onward transportation to dispersed resort areas. Public transit options from the airport and within tourist corridors are limited and poorly advertised, and information on routes, fares, and schedules is often unclear or unavailable in multiple languages. The island's rugged terrain and limited road network constrain route design and service frequency, while the distance between the airport, resort zones, and Phuket Town increases reliance on door-to-door services. Transit provision is fragmented among private operators, and regulatory restrictions prevent buses from stopping directly at major hotels, placing them at a competitive disadvantage relative to taxis and for-hire vehicles. In addition, informal and semi-formal services—such as motorbike taxis—play an important role in meeting tourist travel demand, further complicating efforts to shift trips to higher-capacity transit modes.

LOCATION	BUSINESS	LEISURE	INTERNATIONAL
PHUKET	A smaller fraction than in Las Vegas, often attending smaller events.	Large share of visitors, often couples but also families, segmented across modest income young adventure seekers and higher income tourists seeking and upscale experience	Around 60% of all visitors to Phuket come from ASEAN nations, Russia, India, China, the UK, and other countries. Among international arrivals, Russian tourists rank first, followed by Indian and Chinese visitors, respectively.[3]
LAS VEGAS	Entropy index of the 14% of all visitors, high income, often attending conventions/ events	Visitors seeking nightlife, casinos, and experiences. Often arriving by car (e.g. from greater Los Angeles) but will often travel by foot once arrived	10% of all visitors.

Table 1: Composition of Tourist Travel, Phuket and Las Vegas based on expert opinion from US-ASCP participants.

Credit: PKCD

METHODS FOR IDENTIFYING, UNDERSTANDING, AND ENGAGING STAKEHOLDERS

The previous section underscored the importance of tailoring research to the specific context of a tourism destination. Doing so requires a ground-up understanding of the needs and behaviors of existing and potential transit users, beginning with strategies to identify and engage relevant stakeholders. This section outlines the methods applied in Phuket to build that understanding. It begins with identifying relevant stakeholders and gathering input into issues and opportunities for expanding transit service. The initial scoping work feeds into a travel survey designed to reach key user subgroups, followed by qualitative techniques used to develop transit user personas as practical shortcuts for interpreting travel needs. The section concludes with a validation phase, in which findings were tested and refined through stakeholder engagement to co-design actionable insights. For each method, we describe appropriate use cases, required expertise, indicative costs, and practical tools and approaches.

Stakeholder Delineation

When to use stakeholder delineation: Context-specific research and recommendations should be informed by the people most closely engaged in the operation of transit and who influence how services are used. The composition of relevant stakeholder can vary significantly across locations as institutional structure, the extent of existing services and nature of competition, and even the geography of the location can affect how stakeholders interact with the transit services. In the case of Phuket, for example, the centralization of many transportation decisions, privatization of transit services, and the challenging topography of the island that separates resorts, airport, and urban center were bound to play important roles in determining how transit operates.

Expertise and information needed: Stakeholder delineation should cover five core groups to ensure that tourism mobility planning reflects regulatory, operational, tourism, local, and community perspectives. These groups include

- Regulatory and Planning Institutions responsible for transport policy, licensing, and infrastructure decisions;
- Local government bodies with authority over land use, public space, and local mobility management;
- Transport Operators and Service Providers, including public, private, and informal or semi-formal services;
- Tourism-sector actors such as hotels, attractions, destination management organizations, and business associations;
- Civil society or non-governmental organizations engaged in sustainability, accessibility, or community development.

These five groups provide a minimum structure that can be applied across tourism destinations, while allowing additional stakeholders to be included where relevant.

Approach: The initial stakeholder list is developed through one of two approaches, depending on the availability of local expertise. Where local experts are available, stakeholder delineation begins by consulting individuals with direct knowledge of the local transport and tourism context, such as staff from local governments, transport agencies, operators, universities, or experienced practitioners. These experts are asked to identify key stakeholders within each of the five groups, to flag actors that play influential roles in practice but may not be visible in formal documentation, and to clarify the roles these stakeholders play within the local community and mobility system. Where local experts are not available, the initial list is constructed using publicly available information, including government websites, operator listings, tourism authority materials, business association directories, and relevant reports, with the aim of identifying at least one relevant actor for each group.

Once the initial stakeholder list is established, each stakeholder is briefly mapped according to their primary role in tourism mobility. The list is treated as provisional and reviewed with early contacts in government, transport operations, and the tourism sector to identify any missing actors. Additional stakeholders are incorporated as they are identified, and the review continues until no further actors are suggested. The finalized stakeholder list is then used to guide survey targeting and subsequent engagement and validation activities.

Cost: The main costs likely to be incurred in developing a network of stakeholders is in organizing meetings. Depending on the location of the research team, travel to meet with individuals, booking of meeting rooms, and other expenses associated with arranging meeting will add to the overall project cost.

Travel Surveys

When to use a travel survey: Travel surveys are designed to capture detailed information on travel behavior that cannot be obtained from other sources. They provide rich, disaggregated data on trip origins and destinations, time-of-day patterns (e.g., peak versus off-peak travel), trip purposes, and traveler demographics. For this reason, travel surveys are best suited to infrastructure investment and transit service planning, including decisions about street expansions, highway development, bus lanes, route alignments, stop locations, and service frequencies. ASEAN cities should prioritize travel surveys ahead of major infrastructure or network decisions where understanding who is traveling, why, and under what constraints is critical.

Mobile phone-derived data are an increasingly common alternative and can offer broad coverage at scale, but they lack the behavioral and demographic detail of travel surveys and often involve significant cost and access limitations (see Table 2).[1]

Expertise and Information Needed: Travel surveys require an understanding of survey research methods. Often a survey firm can be engaged to conduct the survey. The sponsoring agency should provide careful oversight, as travel surveys are longer and more complicated than many other surveys. A typical travel survey includes a travel diary, querying the respondent about trips made in a day, trip by trip, asking about the origin, destination, mode of travel, trip purpose, and trip start and end time. This requires that the surveyor be familiar with definitions of what constitutes a trip (e.g., how long of a walk before we call it a trip), and ways to help respondents understand the response choices for travel mode and trip purpose. Often times a travel survey will be stratified, either by income, or gender, or residential location, or other traveler characteristics. Hence the survey research firm needs data on the underlying population (e.g., by income or other characteristics that will be used to draw a stratified sample) and needs to be skilled in survey techniques for stratification. The sponsoring agency – usually a transportation

Box 2. Lessons from Phuket travel survey

We began by setting the main objective: to understand how to encourage a shift toward greater use of public transport along the Phuket–Patong route, where EV bus options are available and can serve both residents and visitors.

We collected background information on the characteristics of the local population and visitors—such as the resident–visitor ratio, nationality, age distribution, and overall mode share in Phuket. Based on this, we designed separate travel survey instruments for residents and visitors.

We determined the required sample size using Cochran, W. G. (1977, p. 75). equation for stated preference survey design, and identified data collection sites within a 500-meter buffer zone, consistent with established evidence on pedestrian catchment areas for transit.

Before full deployment, we conducted a pilot survey with 10% of the sample size to test the instruments and refine the approach. Depending on the target sample size, it is not necessary to always have a pilot that is 10% of the full sample. The purpose of a pilot survey is to test the clarity of questions, to confirm the training of the survey team, to verify that respondents will be willing to participate, and to uncover any unanticipated difficulties. Typically, pilots on the order of 50 to 100 respondents are sufficient to identify these issues.


Credit: Baff

planning or land development agency – needs to have the project management skills to vet, select, and manage the survey contractor, and an understanding of survey methods and basic statistics to use and interpret the results.

Approach

- Set objectives
  - Define what information is needed and why (e.g., trip purpose, mode choice, willingness-to-pay).

DIMENSION	TRAVEL SURVEY	MOBILE PHONE DATA
Primary Data Type	Self-reported travel behavior and perceptions	Passively collected location and movement data
Spatial Coverage (Where)	Limited to survey locations and sampling design	Very high, covering large areas and multiple corridors
Temporal Coverage (When)	Snapshot or short survey period; can be stratified by time	Continuous or near-continuous over long periods
User Identification (Who)	Detailed socio-demographics (resident vs. tourist, age, income, vehicle ownership)	No direct socio-demographic information
Attitudes & Perceptions	Captures satisfaction, preferences, barriers, and willingness to switch modes	Not observable
Sample Size	Moderate to small (resource-dependent)	Very large
Bias Risks	Sampling bias, recall bias, response bias	Representation bias, opaque sam-pling mechanisms
Cost & Implementation Effort	Relatively high (design, fieldwork, incentives)	High data acquisition cost, but low field effort

Table 2: Comparison of travel survey and mobile-phone derived data.

- Design the survey
    - Draft and pretest questions
    - Decide on format (on-board, at/around stops, online follow-up).
- Plan sampling and logistics
    - Identify where to reach respondents (covering both access and egress zones).
    - Recruit and train enumerators.
- Conduct the survey
    - Pilot first, analyze the pilot results, adjust for missing coverage, then implement the full survey.
    - Monitor fieldwork to maintain quality.
- Analyze and report
    - Clean and analyze data.
    - Summarize findings, note limitations, and document methods for future use.

Cost: The travel survey in along the PKCD bus route in Phuket cost approximately $20,000 including $4,000 for survey design and development, $8,000 for the survey fee and $8,000 for the incentive for the respondents) . Generally, travel survey-related costs fall into two main categories:

First, the development of the survey questions, which could be done either in-house by agency staff or in consultation with travel behavior experts. For first-time applications, we advise engaging academic experts skilled in travel surveys to learn from decades of experience in the literature.[2]

Second, survey administration, which is best conducted by an experienced survey research team. Each city will have its own unique context for reaching survey respondents, and skilled local survey research firms will typically best understand that context.

Persona Development

When to use personas: Traveler personas are useful tools to understand traveler motivations from a marketing and consumer satisfaction perspective. Personas are deep profiles, developed from quantitative (e.g., survey) and qualitative (e.g., interview or observational) data, offering a deep understanding of traveler behavior by defining key segments, including their mobility challenges, needs, and opportunities.

Expertise and Information Needed:
Personas require two kinds of expertise:

(1) Skill with qualitative and quantitative methods, often from a marketing or consumer satisfaction perspective, and (2) Knowledge of the local travel customer base. Interviews with experts in transit agencies, bus companies, tourism development firms, and tourist service providers (hotels, restaurants) can be a starting point for understanding the customer base and how to engage in direct customer (traveler) interviews.

Approach:How to develop travel personas:

Clean the survey data: Address any missing or incorrect answers to make sure the information is accurate and ready to use.

Select key characteristics: Choose the most important traits that help distinguish different groups of travelers. For example, travel cost, age, job type, preferred travel mode, trip purpose, or attitudes toward safety and the environment. Ideally, this is done using a mixed method approach based on survey results and qualitative field work. Field work may involve more in-depth interviews with a subset of survey respondents and/or participant observation to note recurring patterns of use.

Group travelers: Use the characteristics identified in Step 2 (above), sort people into groups with similar patterns using statistical analysis. This helps illuminate clear profiles, such as "students who care about safety and environmental issues. (occupation and attitude)"

Define traveler personas: For each group, identify the "typical" traveler who best represents that segment. These personas then serve as realistic examples of the different kinds of travelers in the system.

Once personas are developed, agencies can further engage with real travelers who match these profiles through in-depth interviews or focus groups, or validate them with local stakeholders. This deeper engagement helps uncover motivations, unmet needs, and expectations that may not be fully captured through surveys alone. The resulting insights support the identification of service gaps, opportunities for innovation, and areas where targeted interventions can meaningfully improve the travel experience. This step provides a critical foundation for developing pragmatic and context-sensitive recommendations.

Cost: Cost will vary based on the solution adopted for the survey. Survey firms are unlikely to provide persona development as well, thus requiring the services of a different set of consultants. In the case of the Phuket case study, the Chulalongkorn team was responsible for both survey design and analysis, which included the persona development. Another factor that will affect cost is the degree of engagement with respondents. More extensive qualitative work can be labor and time intensive and, therefore, costs more.

Co-design Recommendations with local stakeholders

When to use: Following the development of traveler personas, the methodology incorporates a co-design process with local stakeholders to translate analytical results into actionable and context-appropriate recommendations. While persona-based clustering provides a data-driven understanding of user and non-user needs, effective policy and service interventions must also reflect local institutional arrangements, operational constraints, and implementation capacity. Engaging stakeholders at this stage ensures that recommendations are both responsive to traveler needs and feasible within the local transport system.

Expertise and Information Needed: The co-design process should involve stakeholders who play key roles across planning, regulation, operation, and service delivery. Typical participants include public transport regulators and planning authorities, public and private bus operators, local government agencies, and tourism-related organizations such as hotel associations or destination management bodies. Where relevant, representatives from business associations or civil society organizations may also be included. This mix of participants helps ensure that multiple perspectives are considered and that proposed actions align with both user expectations and delivery responsibilities.

Approach: The workshop is designed to validate the personas developed in Section B and to co-develop service improvement measures with stakeholders. It typically begins with a presentation of the travel survey findings, followed by an overview of the personas and the associated service gaps identified through the analysis. Stakeholders are then invited to assess whether these profiles align with observed travel behavior and operational experience, and to propose refinements where needed.

To encourage broad participation and maintain engagement, interactive tools may be used, depending on context and availability. Online polling and visualization platforms can support real-time feedback, enable anonymous input, and help surface areas of consensus or divergence among participants. These tools are particularly useful in mixed groups where participants may have different levels of authority or familiarity with the topic.

Following validation, stakeholders work collaboratively to propose interventions that respond to persona-specific needs and cross-cutting service gaps. Facilitation should emphasize practicality, encouraging participants to consider regulatory authority, operational feasibility, and resource implications when proposing actions.

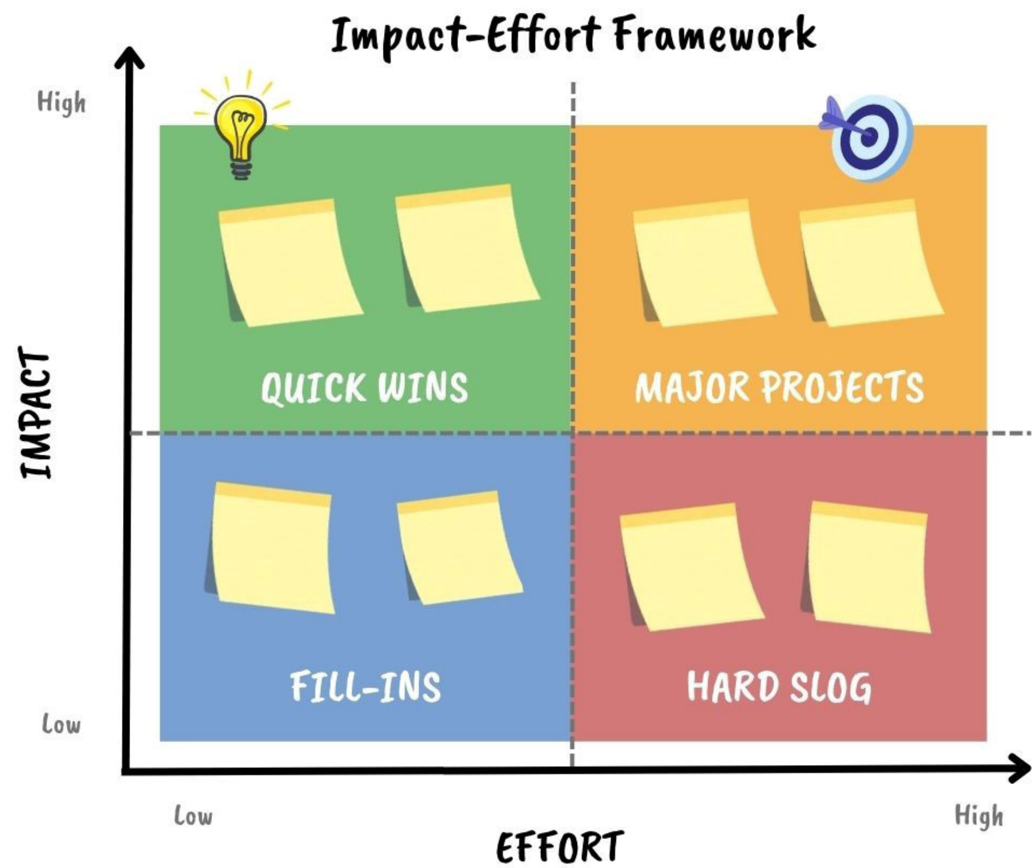

Figure 1 Impact-Effort Framework

To translate a wide range of co-designed ideas into a focused and actionable set of recommendations, the workshop applies an impact–effort framework. This simple and intuitive tool helps stakeholders jointly assess which interventions should be prioritized, sequenced, or set aside.

Proposed actions are assessed along two dimensions:

- Impact referring to the expected contribution to improved service quality, user experience, or ridership
- Effort reflecting the level of resources, coordination, time, and institutional complexity required for implementation.

During the workshop, stakeholders collectively discuss and position each proposed intervention on a two-axis matrix. This process encourages participants to consider not only what would be desirable from a user perspective, but also what is realistic given local operational and institutional constraints.

The resulting matrix typically identifies four broad categories:

- High-impact, low-effort actions, which are suitable for near-term implementation and can deliver visible improvements quickly
- High-impact, high-effort actions, which represent strategic priorities requiring longer-term planning and coordination
- Low-impact, low-effort actions, which may be implemented opportunistically
- Low-impact, high-effort actions, which are generally deprioritized.

Cost: TThe co-design workshop conducted as part of the PKCD bus corridor study in Phuket cost approximately $5,000, including $1,500 for pre-workshop preparation and analysis, and $3,500 for direct workshop implementation. Pre-workshop costs covered the synthesis of travel survey results and persona profiles, preparation of workshop materials, facilitation design, and coordination with participating agencies. Direct workshop costs included venue and logistical arrangements, facilitation materials, and participant-related expenses. More generally, co-design–related costs typically fall into two main categories.

First, pre-workshop preparation, which may be undertaken in-house by agency staff or in collaboration with external researchers or facilitators. This phase includes analytical synthesis, development of personas and preliminary service-gap summaries, workshop design, and stakeholder outreach. For first-time applications, engaging researchers or practitioners experienced in participatory and co-design methods is strongly recommended, as careful preparation is critical to ensuring productive discussion and actionable outcomes.

Second, orkshop administration and implementation, which involves the delivery of the co-design session itself. Depending on local context, this may include venue rental, facilitation and moderation, audio-visual or digital collaboration tools, refreshments, and participant travel or accommodation. As with survey implementation, local institutional arrangements and stakeholder availability strongly influence both the format and cost of co-design activities.

RESULTS FROM A CASE STUDY IN PHUKET, THAILAND

Governance Roles

In Phuket, many of the most consequential regulations and standards are determined at the national level. Accordingly, understanding governance in Phuket requires first clarifying how national level institutions shape the scope of action available to local governments and operators, and how these arrangements are translated into local level implementation outcomes.

Regulatory and Planning Institutions

At the national level, regulatory and planning authority for public transport in Phuket remains largely centralized under the Ministry of Transport. Key institutions include:

- Department of Land Transport (DLT): Responsible for public bus route planning, operator licensing, and regulatory oversight.
- Office of Transport Policy and Planning (OTP): Responsible for national public transport policy formulation and strategic planning.
- Commission for the Management of Land Traffic (CMLT): Provides inter-agency coordination on land traffic management, enforcement alignment, and resolution of operational issues affecting routes and service delivery.

In addition, digital and smart cities initiatives are supported at the national level by the Ministry of Digital Economy:

- Digital Economy Promotion Administration (DEPA): Promotes smart city and smart mobility technologies, including digital platforms and data-driven solutions relevant to tourism mobility.

Local Government Bodies

Provincial and local governments play a coordinating and implementation-oriented role, bridging national policy with local conditions:

- Phuket Provincial Governor's Office: Coordinates relevant public agencies and supports alignment across sectors.
- Provincial Subcommittee of CMLT: Acts as a provincial-level coordination mechanism for traffic and transport issues.
- Phuket Department of Land Transport (Phuket DLT): Serves as the provincial implementation unit of the Department of Land Transport, responsible for applying and enforcing national public transport policies in Phuket context.

Transport Operators and Service Providers

Transport operators and service providers are responsible for translating policy objectives and regulatory frameworks into day-to-day service delivery. In Phuket, service provision reflects a mix of private initiative and local-market responsiveness, particularly in corridors serving high tourist demand.

- Phuket City Development (PKCD): Acts as the parent company of the electric bus operator, Phuket Smart Bus, providing premium-quality public transport services along key tourist corridors. PKCD plays an important role in piloting new service models, fleet electrification, and service branding aimed at reducing reliance on private and rental cars.
- Phuket Mahanakorn: Operates local bus services that primarily serve residents while also accommodating tourist travel needs. The operator plays a critical role in ensuring basic network coverage, affordability, and integration with existing local mobility patterns.

Tourism-Sector Actor

Tourism-sector actors influence travel demand patterns and visitor mobility choices, and therefore play an important enabling role in tourism mobility governance. Their involvement helps align transport services with tourism activity locations, peak periods, and visitor expectations.

- Patong Hotel Association: Represents hotels in Patong, one of Phuket's primary tourist destinations, and acts as a coordination platform between accommodation providers and transport stakeholders. The association supports information dissemination, service feedback, and promotion of public transport options to visitors.
- Thai Hotels Association – Southern Chapter: Represents hotels across southern Thailand, including Phuket, and contributes a broader regional tourism perspective. The association plays a role in advocating for improved accessibility, sustainable transport options, and policies that enhance the overall visitor experience.

Civil Society and Non-Governmental Organization

Civil society and non-governmental organizations contribute to governance by representing broader social, economic, and environmental interests. Their involvement supports accountability, inclusiveness, and alignment between mobility interventions and local development objectives.

- Sustainable Tourism Development Foundation: A non-political foundation dedicated to advancing sustainable tourism and environmental preservation through research, policy support, capacity building, and information dissemination.
- Federation of Thai Industries (FTI): Represents private-sector and industrial interests, including those related to transport, energy, and vehicle manufacturing. In the context of tourism mobility, FTI provides perspectives on economic viability, investment conditions, and the role of electrification and domestic industries in supporting sustainable transport transitions.

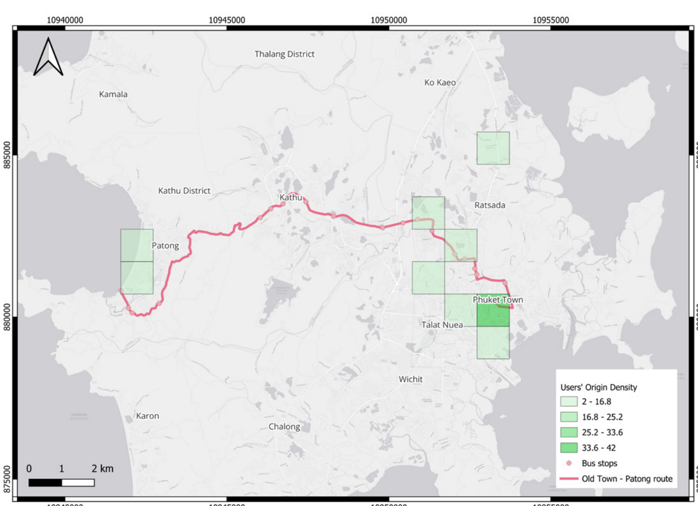

Figure 2. Users' origin

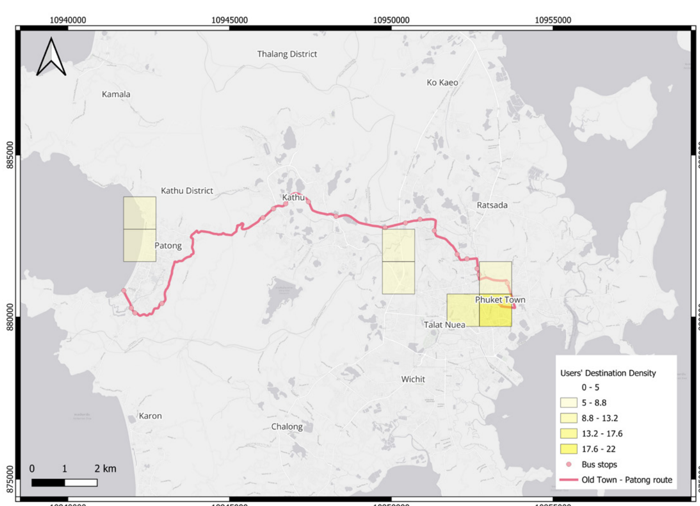

Figure 3. Users' destinations

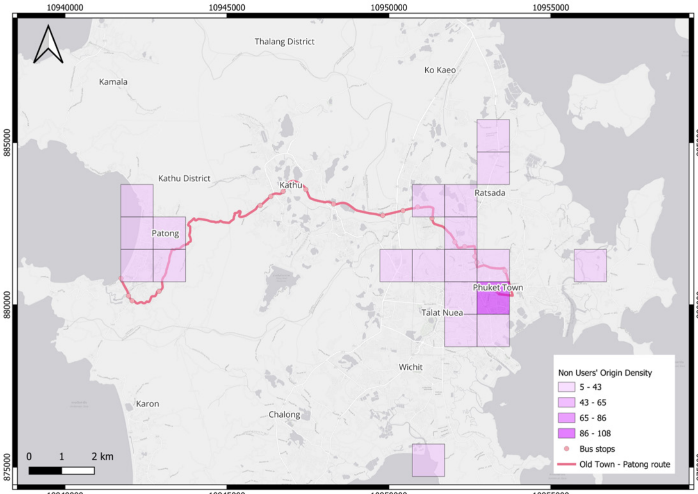

Figure 4. Non-users' origins

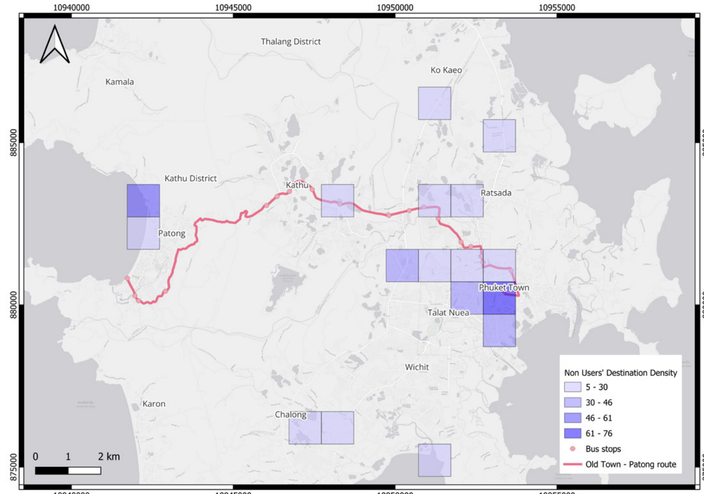

Figure 5. Non-users' destinations

Potential Demand Heatmap (Origin-Destination Density)

Based on the travel survey, we developed a practical tool to generate potential demand heatmaps, showing where people are most likely to start and end their trips along and around the PKCD EV bus corridor. The maps are built using a 1 km × 1 km grid and combine inputs from both current users and non-users, providing a comprehensive view of both existing ridership and latent demand.

The heatmaps highlight areas with higher concentrations of travel activity, where darker shading indicates stronger potential demand. By viewing origins and destinations together, stakeholders can quickly assess whether the current route reflects how people move around the city, and where adjustments may be needed to better match observed travel patterns.

The origin–destination density maps show that both users and non-users are strongly focused on the Phuket Town–Kathu–Patong corridor. Users are most concentrated in Phuket Town (including Talat Nuea) and Patong, which matches the main service corridor. Non-users show a similar corridor pattern, but their destinations also extend beyond the corridor, with additional clusters in Chalong and in the eastern parts of Phuket. Compared with users, non-user destinations are therefore more spatially spread, especially toward areas that are less directly covered by the current route.

Persona-Based Segmentation Results

Using clustering analysis of survey data, the study identified eight distinct personas: four among current bus users and four among non-users shown in Tables 3 and 4. The personas reveal substantial heterogeneity in socio-demographic characteristics, travel behavior, and service expectations within both groups. The segmentation demonstrates that barriers to adoption, travel attitude, and drivers of satisfaction vary systematically across subgroups.

Across all eight personas, six cross-cutting service themes repeatedly emerged as critical determinants of bus use and perceived service quality: (1) Driver professionalism and communication, (2) Waiting time, (3) Integration with mobile applications, (4) Tourist information at stops and on board, (5) Reliability and reduced planning uncertainty,

CLUSTER	PERSONA DESCRIPTION	AGE	GENDER	RESIDENT: VISITOR	FACTORS INFLUENCING CHOICE	COMMON NEEDS
1	Low-income students or workers prioritizing affordability	<30	M:F ≈ 50:50	100:0	Affordability, road safety	Clearer driver communication, better on-board service, stops closer to destination
2	Bus-friendly low–mid income freelancers	18–39	M:F ≈ 24:76	≈37:63	Convenience at start of trip	Professional drivers, shorter waiting time
3	Low–mid income young male visitors interested in adventure	<30	M:F = 100:0	≈2:98	Convenient and affordable transport	Professional drivers, shorter waiting time
4	High-income visitors open to buses	30–39	M:F ≈ 27:73	0:100	Feeling safe and secure	Professional drivers, tourism information on board & at stops

Table. 3. User ersonas Results

CLUSTER	PERSONA DESCRIPTION	AGE	GENDER	RESIDENT: VISITOR	CURRENT MODE	FACTORS FOR MODE CHOICE	KEY NEEDS TO SHIFT TO EV BUS
5	Low-income workers/ students or budget young visitors	<30	M:F ≈ 47:53	48:52	Private MTB, rental MTB	Short wait time, convenience	Cleanliness, app integration
6	Low–mid income freelancers	18–59	M:F ≈ 48:52	58:42	Private MTB/ car, ride-hailing	Convenience, flexibility	App integration, stops closer to destinations, professional drivers
7	Mid income private riders	18–29	M:F ≈ 42:58	≈92:8	Private car/ MTB	Convenience, economy, safety	Driver communication, on-board service, app integration
8	Premium, app-first high-income visitors	18–29	M:F ≈ 42:58	≈5:95	Taxi, ride-hailing	Comfort, convenience, predictability	Cleanliness, punctuality, professional drivers, app integration

Table 4. Non-user personas results

and (6) Cleanliness and comfort. These themes formed the basis for the stakeholder workshop that followed, which was designed to validate the personas and translate persona needs into implementable recommendations.

To ensure the segmentation results were operationally credible and actionable, the research team convened a stakeholder workshop to validate the personas and co-design service improvement measures. The workshop created a structured forum for the following stakeholders:

- Department of Land Transport, Phuket as a regulator
- Phuket City Development as an EV bus operator
- Phuket Mahanakorn as a local bus operator
- Phuket Provincial Administration Organization (PAO) as a local governmental organization.
- Federation of Thai Industries as a local non-governmental organization
- Patong Hotel Association, as a tourism-related organization
- Thai Hotels Association, Southern Chapter, as a tourism-related organization

To confirm whether the personas reflected observed mobility patterns and to prioritize interventions based on feasibility and expected benefits. Following validation, stakeholders worked collaboratively to convert persona-specific needs into concrete interventions. The discussion focused on those repeatedly emerged service theme while remaining realistic about institutional capacity and implementation constraints.

In total, stakeholders consolidated fifteen recommendations, spanning both near-term operational measures and longer-term system reforms. The recommendations cover improvements in boarding and alighting convenience, service frequency and scheduling, timetable transparency, driver training and customer service, digital integration and

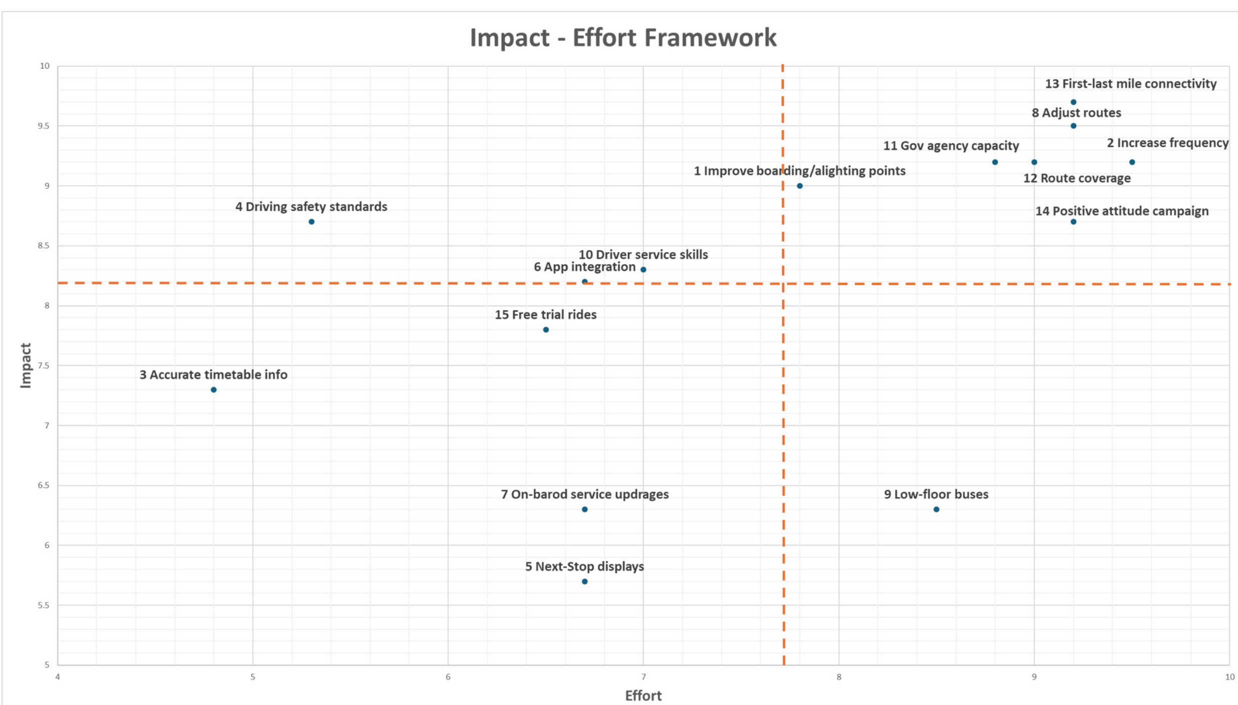

Figure 6. Impact-effort framework

real-time information, vehicle quality and accessibility, route design and coverage, first–last mile connectivity, public awareness campaigns, and trial or promotional fares to encourage initial ridership.

All recommendations were assessed using an impact–effort framework as shown in Figure 6, enabling stakeholders to distinguish between measures that can be implemented quickly and those requiring greater coordination, investment, or structural change. The assessment highlighted a set of quick wins (Top left quadrant)—such as Improve driver capacity and customer-service skills, strengthening driving safety standards and improving digital information through app integration. It also identified several major projects (Top right quadrant) requiring sustained institutional collaboration, including route restructuring, frequency increases, strengthened governmental organization planning capacity, and improved first–last mile connectivity. The resulting prioritization (Figure 6) supported a shared understanding of sequencing: actions that can deliver immediate user-facing improvements versus reforms that enable longer-term system transformation.

LESSONS LEARNED AND RECOMMENDATIONS

Stakeholder Delineation

A key lesson from the stakeholder delineation process is the value of systematically identifying relevant actors at an early stage of tourism mobility planning. Using a clear set of stakeholder groups helped ensure balanced coverage across regulatory, operational, tourism, local government, and civil society perspectives, and reduced the risk of overlooking actors with influence over decision-making or implementation. Engagement with local experts was particularly important for identifying informal roles and practical dynamics that were not evident from formal documentation alone.

A second lesson is the importance of treating stakeholder delineation as an iterative process rather than a one-time exercise. Reviewing the initial stakeholder list with early contacts helped surface additional actors and refine understanding of stakeholder roles. This iterative approach strengthened the alignment between stakeholder engagement, survey targeting, and subsequent validation activities, and improved the practical relevance of the findings.

Travel Survey

A key lesson from the travel survey is that careful survey design and rigorous data cleaning are essential for producing reliable geographic insights and supporting persona development. Decisions about where, when, and from whom data are collected directly affect the quality of origin and destination density grids and the conclusions drawn from them. In a tourism-oriented context such as Phuket, balanced sampling across users and non-users, locations, and time periods is especially important to reduce bias from convenience-based intercept surveys. In addition, attitudinal questions should use a clear and consistent scale direction to avoid confusion and ensure that responses can be interpreted and compared correctly.

Furthermore, spatial analysis provides a clear picture of where demand is concentrated. The maps indicate that both users and non-users are strongly focused on the Phuket Town-Kathu-Patong corridor, meaning that many non-users already travel to destinations served by the route. This suggests that non-use along the corridor is less about destination mismatch and more about service performance compared with other modes. Importantly, the presence of non-user destinations within the service corridor signals practical potential demand and helps define clear, stakeholder-focused themes for action, such as improving reliability, service frequency, and passenger information. These themes were then carried forward into the workshop co-design process to support the development and prioritization of stakeholder recommendations.

Persona Development

A key lesson from the persona development process is that understanding travel behavior and service needs requires going beyond demographic characteristics alone. While variables such as age, income, or occupation provide useful background, they do not fully explain why people choose to use or avoid public transport. By combining behavioral and attitudinal information, the personas capture differences in travel habits, confidence levels, and expectations toward the service, offering a more complete picture of both users and non-users.

The Phuket personas further indicate that public transport attractiveness in tourism-oriented contexts depends not only on demographic characteristics, but critically on the overall service experience. Predictability, driver professionalism, clarity of information, and perceived ease of use play a central role in shaping confidence among both residents and visitors, including those who do not currently use bus services. At the same time, the analysis reveals that

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several needs are shared across different personas, despite differences in socio-economic profiles. Common concerns related to reliability, ease of use, and information quality appear repeatedly across groups.

Co-Design recommendation with stakeholders

A key lesson from the co-design phase is that analytical outputs become significantly more valuable when they are validated and refined with local stakeholders. While personas help clarify who different traveler groups are and what they need, stakeholders are essential for confirming whether these insights reflect local realities and whether proposed actions are feasible given operational constraints, budget limitations, and institutional roles. In Phuket, engaging regulators, operators, local agencies, and tourism-sector actors ensured that personas and service gaps were treated not as standalone research findings, but as practical inputs for decision-making.

A second lesson is that structured prioritization is necessary to translate a broad set of ideas into an actionable plan. Applying an impact-effort framework provided stakeholders with a shared basis to compare options and agree on near-term priorities, particularly under real-world constraints such as limited funding, staffing capacity, route licensing requirements, and operational feasibility. This approach supports clearer sequencing of actions and helps align improvement ambitions with implementation capacity.

To further development, evidence and prioritized recommendations will not deliver change on their own without a clear mechanism to carry them forward. While research teams can generate analysis and convene evidence-based dialogue, sustained implementation typically requires a multi-stakeholder structure. In practice, public agencies provide policy direction, regulatory oversight, and coordination, while transit operators and tourism-sector partners translate agreed priorities into service design changes and customer-facing improvements. The core challenge, therefore, is not only selecting the right interventions, but also establishing a credible lead or host institution with the mandate and capacity to steward the process from research to implementation.

Lessons from Governance Roles

A key lesson from the Phuket case is that misalignment between governance roles can substantially constrain public transport performance in tourism-oriented cities. While bus planning and regulatory authority remain centralized at the national level, responsibility for operations and financing is largely delegated to local governments and private-sector actors. Stakeholder engagement revealed that limited technical capacity and resources within central agencies, combined with constrained implementation capacity at the local level, contributed to frequent delays or cancellations of planned services.

A second lesson concerns the role of private-sector actors in filling institutional gaps. In Phuket, the emergence of Phuket Smart Bus as a privately led initiative reflected unmet demand for higher-quality public transport that was not being addressed through conventional public-sector mechanisms. Collaboration with the Department of Land Transport enabled the launch of a new service between Phuket Town and Patong in October 2024, demonstrating the potential of hybrid governance arrangements.

However, a third lesson is that institutional collaboration alone is insufficient without adequate data and demand intelligence. Stakeholder discussions highlighted limited availability of reliable travel demand data and insufficient understanding of heterogeneous passenger needs, particularly differences between tourists and local residents. As a result, service design was not fully aligned with underlying travel patterns, contributing to lower-than-expected ridership levels.

ENDNOTES

1 Alexander, L., Jiang, S., Murga, M., & González, M. C. (2015) "Origin–destination trips by purpose and time of day inferred from mobile phone data." Transportation Research Part C: Emerging Technologies, 58, 240–250. Toole, J. L., Colak, S., Sturt, B., Alexander, L. P., Evsukoff, A., & González, M. C. (2015) "The path most traveled: Travel demand estimation using big data resources." Transportation Research Part C: Emerging Technologies, 58, 162–177

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